

EE227 – Digital Logic Design

Lecture 24

Outline

- Design of Sequential Circuits
 - Design with D Flip Flop
 - Excitation Table
 - Design with JK Flip Flop
 - Design with T Flip Flop

Design of Sequential Circuits

- The design of a clocked sequential circuit starts from a set of specifications/word description and culminates in a logic diagram or a list of Boolean functions from which the logic diagram can be obtained
- A synchronous sequential circuit is made up of flip-flops and combinational gates
 - The design of the circuit consists of choosing the flip-flops and then finding a combinational gate structure that, together with the flip-flops, produces a circuit which fulfills the stated specifications
- The number of flip-flops is determined from the number of states needed in the circuit
- The combinational circuit is derived from the state table by evaluating the flip-flop input equations and output equations

Steps to Design a Sequential Circuit

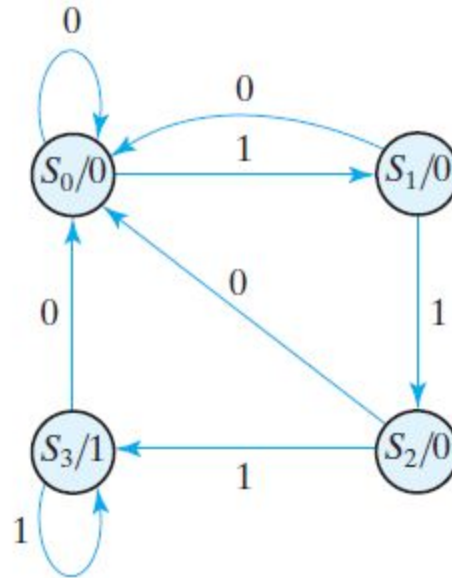
- The procedure for designing synchronous sequential circuits can be summarized by a list of recommended steps:
 - 1) From the word description and specifications of the desired operation, derive a state diagram for the circuit
 - 2) Reduce the number of states if necessary
 - 3) Assign binary values to the states
 - 4) Obtain the binary-coded state table
 - 5) Choose the type of flip-flops to be used
 - 6) Derive the simplified flip-flop input equations and output equations
 - 7) Draw the logic diagram

Example 1

- Design a sequential circuit that detects a sequence of three or more consecutive 1's in a string of bits coming through an input line (in other words design a sequence detector)
- Solution
 - Whenever a 0 is at input return to state S_0 and output is 0
 - When a 1 is at input move to next state (S_0 to S_1 , S_1 to S_2 , S_2 to S_3)
 - The output is 1 when three consecutive 1's are detected
 - If more 1's after the three consecutive 1's then remain on the same state (S_3 in this case) and output is 1

Example 1 (Solution)

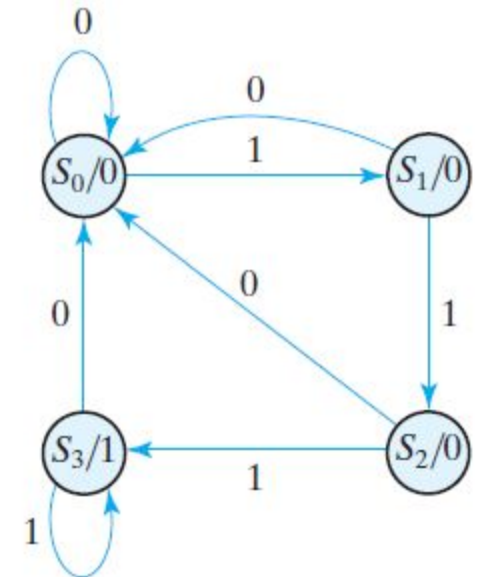
- State Diagram



Example 1 (Solution)

- Design with D Flip Flop
 - The next state and Flip Flop input equation are same for D flip flop
 - For 4 states we need 2 D flip flops
 - 1 input and 1 output

Present State		Input	Next State		Output
A	B		A	B	
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	0
0	1	1	1	0	0
1	0	0	0	0	0
1	0	1	1	1	0
1	1	0	0	0	1
1	1	1	1	1	1



Example 1 (Solution)

- Design with D Flip Flop
 - Flip Flop input equations and output equation
 - Obtained by using k – map

Present State		Input	Next State		Output
A	B	x	A	B	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	0
0	1	1	1	0	0
1	0	0	0	0	0
1	0	1	1	1	0
1	1	0	0	0	1
1	1	1	1	1	1

A \ Bx				
	00	01	11	10
0	m ₀	m ₁	m ₃ 1	m ₂
1	m ₄	m ₅ 1	m ₇ 1	m ₆

$$D_A = Ax + Bx$$

A \ Bx				
	00	01	11	10
0	m ₀	m ₁ 1	m ₃	m ₂
1	m ₄	m ₅ 1	m ₇ 1	m ₆

$$D_B = Ax + B'x$$

A \ Bx				
	00	01	11	10
0	m ₀	m ₁	m ₃	m ₂
1	m ₄	m ₅	m ₇ 1	m ₆ 1

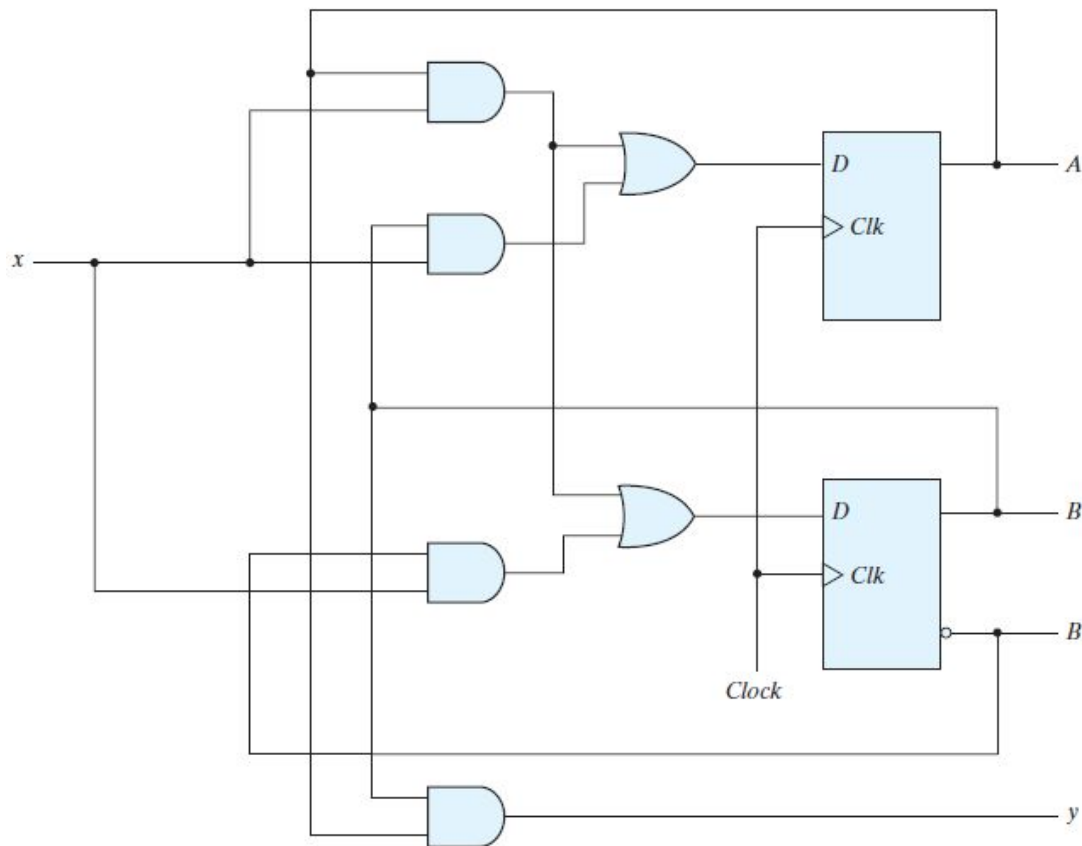
$$y = AB$$

Example 1 (Solution)

- Design with D Flip Flop

- Circuit Diagram

$$D_A = Ax + Bx \quad D_B = Ax + B'x \quad y = AB$$



Design with JK and T Flip Flop

- The design of a sequential circuit with flip-flops other than the D type is complicated by the fact that the input equations for the circuit must be derived indirectly from the state table
- When D -type flip-flops are employed, the input equations are obtained directly from the next state
- This is not the case for the JK and T types of flip-flops
- In order to determine the input equations for these flip-flops, it is necessary to derive a functional relationship between the state table and the input equations
- This is done with the help of excitation tables

Excitation Table

- A table that lists the required inputs for a given change of state is called an excitation table

Flip-Flop Excitation Tables

$Q(t)$	$Q(t = 1)$	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

JK Flip-Flop

$Q(t)$	$Q(t = 1)$	T
0	0	0
0	1	1
1	0	1
1	1	0

T Flip-Flop

Design with JK Flip Flop (1/3)

- Example 2
 - Design a sequential circuit given by the following state table
- Solution
 - State table is obtained with the help of excitation table of JK flip Flop

Present State		Input	Next State	
A	B		A	B
0	0	0	0	0
0	0	1	0	1
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

Present State		Input	Next State		Flip-Flop Inputs			
A	B		A	B	J_A	K_A	J_B	K_B
0	0	0	0	0	0	X	0	X
0	0	1	0	1	0	X	1	X
0	1	0	1	0	1	X	X	1
0	1	1	0	1	0	X	X	0
1	0	0	1	0	X	0	0	X
1	0	1	1	1	X	0	1	X
1	1	0	1	1	X	0	X	0
1	1	1	0	0	X	1	X	1

$Q(t)$	$Q(t = 1)$	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Design with JK Flip Flop (2/3)

- Flip Flop Input Equations
 - Obtained by using K – Map

		Bx			
		00	01	11	10
A	0	m_0	m_1	m_3	m_2 1
	1	m_4 X	m_5 X	m_7 X	m_6 X

$$J_A = Bx'$$

		Bx			
		00	01	11	10
A	0	m_0 X	m_1 X	m_3 X	m_2 X
	1	m_4	m_5	m_7 1	m_6

$$K_A = Bx$$

		Bx			
		00	01	11	10
A	0	m_0	m_1 1	m_3 X	m_2 X
	1	m_4	m_5 1	m_7 X	m_6 X

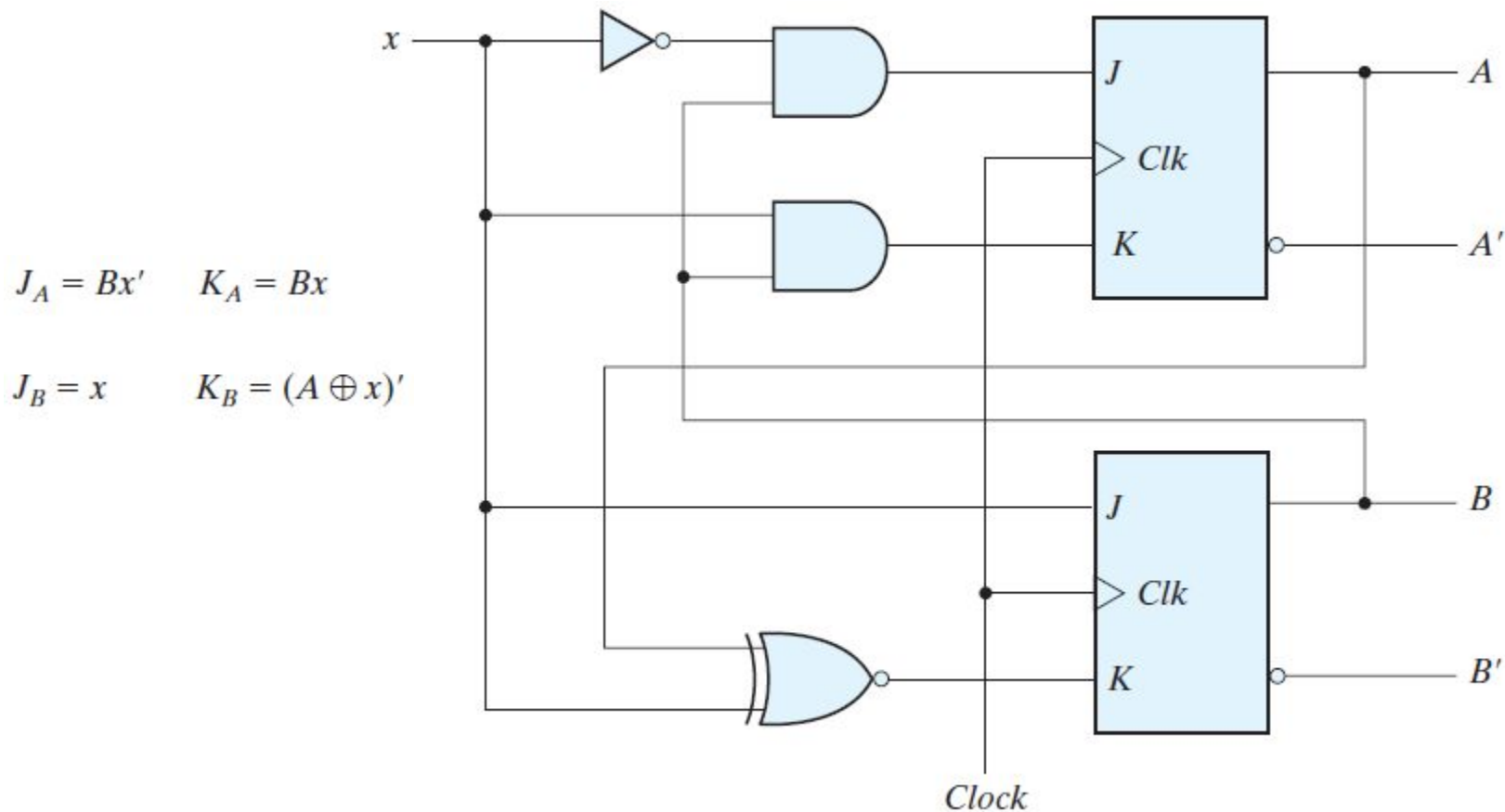
$$J_B = x$$

		Bx			
		00	01	11	10
A	0	m_0 X	m_1 X	m_3	m_2 1
	1	m_4 X	m_5 X	m_7 1	m_6

$$K_B = (A \oplus x)'$$

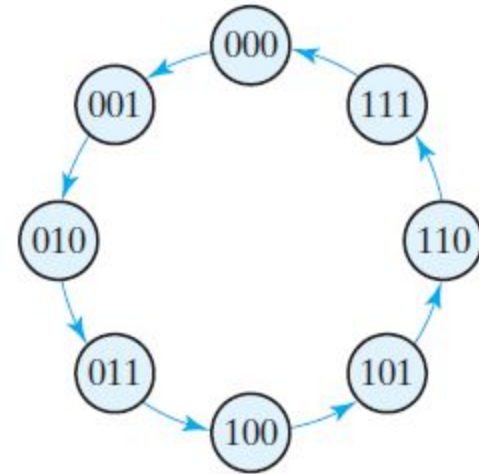
Design with JK Flip Flop (2/3)

- Circuit Diagram



Design with T – Flip Flop (1/4)

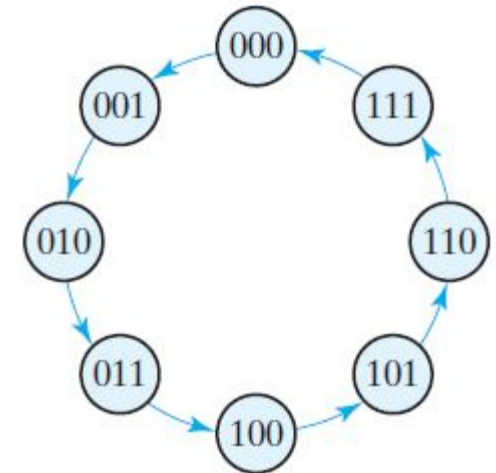
- Example 3
 - Design a 3 bit binary counter
- Solution
 - No input is required
 - No output is required
 - At every clock pulse the circuit should move to next state
 - 3 bit counter, so 8 states are needed (can count from 0 to 7)
 - State diagram is given above



Design with T – Flip Flop (2/4)

- State Table
 - State table is obtained with the help of excitation table of T flip flop

Present State			Next State			Flip-Flop Inputs		
A_2	A_1	A_0	A_2	A_1	A_0	T_{A2}	T_{A1}	T_{A0}
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	1	0	1	1	1	0	1	1
1	1	1	0	0	0	1	1	1



$Q(t)$	$Q(t = 1)$	T
0	0	0
0	1	1
1	0	1
1	1	0

Design with T – Flip Flop (3/4)

- Flip Flop Input Equations
 - Obtained by using K – Map

Present State			Next State			Flip-Flop Inputs		
A_2	A_1	A_0	A_2	A_1	A_0	T_{A2}	T_{A1}	T_{A0}
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	1	0	1	1	1	0	1	1
1	1	1	0	0	0	1	1	1

A_2	A_1A_0			
	00	01	11	10
0	m_0	m_1	m_3 1	m_2
1	m_4	m_5	m_7 1	m_6

$$T_{A2} = A_1A_0$$

A_2	A_1A_0			
	00	01	11	10
0	m_0	m_1 1	m_3 1	m_2
1	m_4	m_5 1	m_7 1	m_6

$$T_{A1} = A_0$$

A_2	A_1A_0			
	00	01	11	10
0	m_0 1	m_1 1	m_3 1	m_2 1
1	m_4 1	m_5 1	m_7 1	m_6 1

$$T_{A0} = 1$$

Design with T – Flip Flop (4/4)

- Circuit Diagram

